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Machine Learning for Network Attacks Classification and Statistical Evaluation of Adversarial Learning Methodologies for Synthetic Data Generation

Iakovos-Christos Zarkadis
University of Piraeus
Athens, Greece
iakovos.zarkadis@gmail.com

Christos Douligeris
Dept. of Informatics
University of Piraeus
Piraeus, Greece
cdoulig@unipi.gr

***Abstract*—Supervised detection of network attacks has always been a critical part of network intrusion detection systems (NIDS). Nowadays, in a pivotal time for artificial intelligence (AI), with even more sophisticated attacks that utilize advanced techniques, such as generative artificial intelligence (GenAI) and reinforcement learning, it has become a vital component if we wish to protect our personal data, which are scattered across the web. In this paper, we address two tasks, in the first unified multi-modal NIDS dataset, which incorporates flow-level data, packet payload information and temporal contextual features, from the reprocessed CIC-IDS-2017, CIC-IoT-2023, UNSW-NB15 and CIC-DDoS-2019, with the same feature space. In the first task we use machine learning (ML) algorithms, with stratified cross validation, in order to prevent network attacks, with stability and reliability. In the second task we use adversarial learning algorithms to generate synthetic data, compare them with the real ones and evaluate their fidelity, utility and privacy using the SDV framework, f-divergences, distinguishability and non-parametric statistical tests. The findings provide stable ML models for intrusion detection and generative models with high fidelity and utility, by combining the Synthetic Data Vault framework, the TRTS and TSTR tests, with non-parametric statistical tests and f-divergence measures.**

***Index Terms*—Machine learning (ML), artificial intelligence (AI), generative artificial intelligence (GenAI), network intrusion detection systems (NIDS), Unified Multimodal Network Intrusion Detection System (UMNIDS), generative adversarial networks (GANs).**

I. Introduction

In the era of AI, detecting the type of the attack becomes even more difficult. Machine learning algorithms have been implemented in many research papers and application areas [10] showing efficiency in tackling attacks across a variety of well-known intrusion detection system (IDS) datasets [38], [24], [29], [30], [32]. Nevertheless, it is necessary to not simply focus on achieving high results in classification metrics, but offering reliability and stability to an intrusion detection system.

In the last few years adversarial attacks have gained popularity due to their sophisticated attacks, that utilize adversarial machine learning algorithms to inject perturbations in the data, in order to make a strong classifier unable to classify certain attacks correctly, thus misleading the IDS.

Many types of algorithms, from linear models and SVM to neural networks and reinforcement learning have been proposed in [5]. Commonly used algorithms in IoT tabular data are logistic regression as a baseline for multi-class classification as well as more advanced models like deep neural networks and ensemble models [9]. More detailed studies have examined cross-dataset validation with LDA, Xgboost and decision trees [24]. Different frameworks have been widely used for data generation purposes like Synthcity [40] and SDV [26], [8] offering high quality generators and evaluating metrics. The attention of researchers has recently been directed in many conditional architectures for synthetic network data generation for GANs and VAEs [20], [21], [22], [25], [27], [31], as well as in the emerging f-GANs and diffusion models [6], [12], [39] with the LLMs being top-tier models [37]. Furthermore, the specific non-parametric statistical tests, for means, covariance matrices and multivariate distribution comparison, have been used in some papers considering network data [14], [17], [19], but may have never been combined to offer a complete view. Our focus, in this paper is not how to perform such attacks but on how the generation process evolves and test different adversarial algorithms, diffusion forests and LLM to create synthetic data [27], [31] and to evaluate which are the most efficient ones, in terms of fidelity, utility and privacy. In this paper we use TRTS, TSTR tests, supervised distinguishability tests, f-divergences, privacy measurement [11] and combine for the first time the SDV framework [34], with robust non-parametric statistical tests for comparing multivariate means, covariance matrices and the feature's joint multivariate distributions [14], [17], [19].

This work is organized in two sections. In the first we present the classification of the attacks, where we discuss data collection, pre-processing steps and the ML models used for intrusion detection. In the second section we show the generation process of synthetic data, including a short review of the mathematical literature, the training of adversarial, diffusion

網路攻擊分類的機器學習與合成數據生成之對抗性學習方法學統計評估

Iakovos-Christos Zarkadis
比雷埃夫斯大學
希臘雅典
iakovos.zarkadis@gmail.com

Christos Douligeris
資訊工程學系
比雷埃夫斯大學
希臘比雷埃夫斯
cdoulig@unipi.gr

摘要—受監督的網路攻擊偵測始終是網路入侵偵測系統 (NIDS) 的關鍵部分。如今，在人工智能 (AI) 的關鍵時期，隨著更複雜的攻擊手段，例如生成式人工智慧 (GenAI) 和強化學習，若我們希望保護散佈在網路上的個人數據，這已經變成了一個至關重要的組成部分。在本文中，我們處理兩個任務，在第一個統一的雜模態 NIDS 數據集中，該數據集包含流量層數據、封包有效載荷信息和時間上下文特徵，來自重新處理的 CIC-IDS-2017、CIC-IoT-2023、UNSW-NB15 和 CIC-DDoS-2019，具有相同的特徵空間。在第一個任務中，我們使用機器學習 (ML) 算法，並採用分層交叉驗證，以防止網路攻擊，確保穩定性和可靠性。在第二個任務中，我們使用對抗學習算法來生成合成數據，將其與真實數據進行比較，並使用 SDV 框架、 f-發散度、可區分性和非參數統計測試來評估其保真度、效用和隱私。研究結果提供了穩定的 ML 模型進行入侵偵測，以及具有高保真度和效用的生成模型，這是通過結合合成數據保險庫框架、TRTS 和 TSTR 測試、非參數統計測試和 f-發散度度量實現的。索引詞—機器學習 (ML)、人工智慧 (AI)、生成式人工智慧 (GenAI)、網路入侵偵測系統 (NIDS)、統一雜模態網路入侵偵測系統 (UMNIDS)、生成對抗網絡 (GANs)。

一、引言

在人工智能时代，检测攻击类型变得更加困难。機器學習演算法已經被實現於許多研究論文和應用領域[10] ，顯示出在處理各種廣為人知的入侵偵測系統 (IDS) 資料集 [38],[24], [29], [30], [32] 方面的效率。然而，我們必須不僅僅關注在分類指標上達到高結果，還要為入侵偵測系統提供可靠性和穩定性。

過去幾年來，對抗性攻擊因其複雜的攻擊方式——利用對抗性機器學習算法在數據中注入干擾——而日益受到關注，目的是讓強勁的分類器無法正確分類某些攻擊，從而誤導入侵檢測系統（IDS）。

許多種類的演算法，從線性模型和 SVM 到神經網路和強化學習，已在 [5] 中被提出。常見的 IoT 表格數據演算法包括邏輯回歸作為多類分類的基線，以及更先進的模型如深度神經網路和集成模型 [9]。更詳細的研究已經檢查了使用 LDA、Xgboost 和決策樹 [24] 的跨資料集驗證。不同的框架已被廣泛使用於數據生成目的，如 Synthcity [40] 和 SDV [26],[8] ，提供高品質的生成器和評估指標。研究者的注意力最近已經集中於許多條件架構，用於生成合成網路數據以供 GANs 和 VAEs [20], [21], [22], [25], [27], [31], 使用，以及正在興起的 f-GANs 和擴散模型 [6], [12], [39]，其中 LLM 是頂級模型 [37]。此外，一些考慮網路數據的論文已經使用過特定的非參數統計測試，如均值、共變異數矩陣和多變量分佈比較，但可能從未將它們結合起來提供完整的視角。在本論文中，我們的重點不是如何執行這樣的攻擊，而是在於生成過程如何演變，並測試不同的對抗演算法、擴散森林和 LLM 以創建合成數據 [27], [31] ，並評估哪些在保真度、效用和隱私方面是最有效的。在本論文中，我們使用 TRTS、TSTR 測試、監督區別性測試、 f-發散、隱私測量[11] ，並首次將 SDV 框架[34],與穩固的非參數統計測試結合起來，用於比較多變量均值、共變異數矩陣和特徵的聯合多變量分佈 [14], [17], [19]。

這份工作分為兩個部分。第一部分我們介紹攻擊的分類，討論數據收集、預處理步驟以及用於入侵偵測的機器學習模型。第二部分我們展示通用-

合成數據的處理過程，包括對數學文獻的簡要回顧、對抗性、擴散

and LLM algorithms and finally the evaluation procedure.

II. Machine Learning-Based NIDS Methodology

A. Data Collection

For all the results that will be presented in the next sections we used the under-sampled version of the UMNIDS dataset, a newly created dataset, which was created by processing raw pcap files from CIC-IDS 2017, CIC-IOT 2023, UNSW-NB15 and CIC-DDoS 2017 [34]. UMNIDS includes flow-level and payload-based data with features that capture historical and temporal patterns in network traffic. This dataset includes around $170,000$ observations, 88 features and 1 target variable, which implies if a record is normal or if it is an attack in the network along with the type of the attack. Some that are included are Worms, Bots, DDoS, Infiltration and SQL-Injection. The target variable includes a total of 44 unique values, 43 of which specify the kind of attack and the one remaining the non-malicious records. One of the main issues with this dataset is the lack of records for some attacks, like Heartbleed and Infiltrations.

B. Data Pre-Processing

The pre-processing procedure started by omitting the duplicate values, then encoding the label variable, testing different methods like binning for certain numeric variables with a small number of unique values. We performed an in-depth statistical analysis that included the discovering of linear and non-linear relationships, hypothesis testing for existence of normality through QQ-Plots and goodness of fit tests like Kolmogorov-Smirnov and Anderson Darling, as well as regression analysis for some numeric features. We also examined the existence of outliers with IQR, Z-score, DBSCAN and Local-Outlier Factor and due the great number of outliers we preferred to keep them and use as a scaling method the Robust scaler, after we performed an $80/20$ train/test stratified split, with respect to the attack type variable.

Furthermore a great deal of feature engineering techniques were used to increase the efficiency of the ML algorithms, separately for the training and testing data. To tackle the problem of the unbalanced dataset, we investigated different over-sampling and under-sampling techniques only for the training set to avoid data leakage, like synthetic minority oversampling (SMOTE), edited nearest neighbor (ENN) and adaptive synthetic sampling (ADASYN) that lead us to select the over-sampling approach to reduce the impact of information loss. Finally for the feature selection procedure we tested multiple selection techniques like recursive feature elimination, stepwise (forward-backward) elimination and boruta.

C. Attacks Classification

For the classification of the tabular data, we selected baseline and top-tier models from different families. The two baseline models that we used for the multi-class classification task are Logistic Regression and the stochastic gradient descent-based SVM. The tree-based model that we utilized is a Decision Tree. The ensemble models are LightGBM, XGBoost, XGBoost with Random Forest, Random Forest, Extra Trees, Gradient Boosting, and Histogram-Based Gradient Boosting. Finally we also chose to use the AutoGluon, an ensemble of neural networks, extra trees, XGBoost, LightGBM and linear models. Several hyperparameter optimization techniques were tested, such as grid-search, halving-grid-search, and Bayesian optimization, along with stratified 10-fold cross validation. After the selection of the hyperparameters, each of the pre-mentioned models has been trained with stratified 10-fold cross validation to observe their behavior without probability calibration. After that, we performed stratified 10-fold cross validation with probability calibration, while holding a 4% value range threshold for considering a model stable per metric, and obtained the results that follow for the training and testing set.

The metrics with which we evaluated the performance of each algorithm are F1, Precision, Recall, and Roc-Auc. From the results in Tab. I and Tab. II we observe the high performance of ensemble and tree-based models, and more specifically of XGBoost, LightGBM, AutoGluon, XGBoost-RF, Random Forest, Extra Trees, and Decision Tree, while the baseline models, SVM and Logistic Regression fail. To highlight the impressive performance of the non-calibrated XGBoost, in terms of both stability and efficiency, we present the cross validation results for precision and recall in Fig. 1.

TABLE I
Train/Test Precision and Recall Results (%)

Model	*Train/Test-Precision*	*Train/Test-Recall*
LightGBM	97.74/96.36	97.69/96.25
AutoGluon	92.34/93.53	91.32/92.30
XGBoost	98.67/96.40	98.64/96.29
XGBoost-RF	96.04/95.09	95.99/94.74
Random Forest	96.64/95.88	96.56/95.63
HB-Gradient Boosting	94.74/94.74	93.91/93.91
Gradient Boosting	71.82/75.17	71.03/73.47
Extra Trees	97.05/96.30	96.91/96.09
Decision Tree	96.92/95.28	96.85/95.03
SVM	7.08.5.57	0.32/3.41
Logistic Regression	8.16/6.40	6.47/6.82

TABLE II
Train/Test F1 and Roc-Auc Results (%)

Model	*Train/Test-F1*	*Train/Test-Roc-Auc*
LightGBM	97.69/96.25	98.81/98.08
AutoGluon	91.12/92.31	95.56/9606
XGBoost	98.65/96.33	99.30/98.11
XGBoost-RF	95.99/94.86	98.95/97.32
Random Forest	95.56/95.49	96.56/95.63
HB-Gradient Boosting	94.66/94.55	97.28/97.18
Gradient Boosting	69.14/72.78	85.18/86.64
Extra Trees	96.92/96.15	98.42/98.00
Decision Tree	96.85/95.11	98.37/97.46
SVM	1.29/1.34	50.45/50.50
Logistic Regression	5.05/4.84	52.15/52.16

III. Generation of Synthetic Data

The generation of synthetic data has gained popularity nowdays, especially with the hype of GenAI. It is very interesting

算法和LLM算法的最終評估程序。

II. 基於機器學習的網路入侵偵測系統方法論

A. 資料收集

針對下一節將呈現的所有結果，我們使用了UMNIDS資料集的欠樣本版本，這是一個新創建的資料集，由處理CIC-IDS 2017、CIC-IOT 2023、UNSW-NB15和CIC-DDoS 2017 [34]的原始pcap檔案所建立。UMNIDS包含流量層級和有效載荷基礎的資料，其特徵能捕捉網路流量中的歷史和時間模式。這個資料集包含約170,000筆觀察值、88個特徵和1個目標變數，目標變數表示記錄是否正常，或是否為網路中的攻擊以及攻擊類型。其中包含的攻擊類型有蠕蟲、機器人、DDoS、入侵和SQL注入。目標變數總共有44個獨特值，其中43個指定攻擊類型，剩下的1個則代表非惡意記錄。這個資料集的主要問題之一是某些攻擊（如Heartbleed和入侵）的記錄不足。

B. 資料預處理

預處理程序開始時，會忽略重複值，然後對標籤變數進行編碼，對具有少量唯一值的某些數值變數測試不同的方法，例如分箱。我們進行了深入的統計分析，包括發現線性和非線性關係、透過QQ圖和Kolmogorov-Smirnov及Anderson Darling等適合度檢驗來檢驗正態性假設，以及對某些數值特徵進行迴歸分析。我們也使用IQR、Z分數、DBSCAN和Local-Outlier Factor檢查異常值的存否，由於異常值數量眾多，我們選擇保留它們，並使用Robust scaler作為縮放方法，在進行 $80/20$訓練/測試分層隨機拆分後，針對攻擊類型變數進行。

此外，為了提高機器學習算法的效率，我們分別對訓練和測試資料使用了大量的特徵工程技術。為了處理不平衡數據集的問題，我們僅對訓練集調查了不同的過度採樣和欠採樣技術以避免數據洩漏，例如合成少數過度採樣（SMOTE）、編輯最近鄰居（ENN）和自適應合成採樣（ADASYN），這讓我們選擇過度採樣方法以減少信息損失的影響。最後，在特徵選擇程序中，我們測試了多種選擇技術，例如遞迴特徵消除、步進（前向-後向）消除和boruta。

C. 攻擊分類

為了對表格數據進行分類，我們選擇了來自不同系列的基準模型和頂級模型。我們用於多類分類任務的兩個基準模型是邏輯回歸和基於隨機梯度下降的SVM。我們使用的樹模型是決策樹。集成模型包括LightGBM、XGBoost、帶有隨機森林的XGBoost、隨機森林、額外樹、梯度提升和基於直方圖的梯度提升。最後，我們還選擇使用AutoGluon，它是一個由神經網絡、額外樹、XGBoost、LightGBM和線性模型組成的集成。我們測試了多種超參數優化技術，例如網格搜索、減半網格搜索和貝葉斯優化，以及分層10折交叉驗證。在選擇超參數後，我們對上述每個模型進行分層10折交叉驗證，以觀察其行為而不進行概率校準。之後，我們進行帶有概率校準的分層10折交叉驗證，同時保持4%的值範圍閾值，以考慮每個模型在每個指標上穩定，並獲得了以下訓練和測試集的結果。

我們評估每個演算法效能的指標有 F1、精確率、召回率與 Roc-Auc。從表 I 和表 II 的結果中，我們觀察到集成與樹狀模型的表現優異，特別是 XGBoost、LightGBM、AutoGluon、XGBoost- RF、隨機森林、額外樹與決策樹，而基線模型 SVM 和邏輯斯迴歸則失敗。為突顯未校準 XGBoost 非常驚人的表現，在穩定性和效率方面，我們在圖 1 中呈現了精確率和召回率的交叉驗證結果。

表 I
訓練/測試 精確率與召回率結果 (%)

模型	訓練/測試-精確率	訓練/測試-召回率
LightGBM	97.74/96.36	97.69/96.25
AutoGluon	92.34/93.53	91.32/92.30
XGBoost	98.67/96.40	98.64/96.29
XGBoost-RF	96.04/95.09	95.99/94.74
隨機森林	96.64/95.88	96.56/95.63
HB-梯度提升	94.74/94.74	93.91/93.91
梯度提升	71.82/75.17	71.03/73.47
Extra Trees	97.05/96.30	96.91/96.09
決策樹	96.92/95.28	96.85/95.03
SVM	7.08.5.57	0.32/3.41
邏輯回歸	8.16/6.40	6.47/6.82

表 II
訓練/測試 F1 與 ROC-AUC 結果 (%)

模型	訓練/測試-*F1*	訓練/測試-*Roc-Auc*
LightGBM	97.69/96.25	98.81/98.08
AutoGluon	91.12/92.31	95.56/9606
XGBoost	98.65/96.33	99.30/98.11
XGBoost-RF	95.99/94.86	98.95/97.32
隨機森林	95.56/95.49	96.56/95.63
HB-梯度提升	94.66/94.55	97.28/97.18
梯度提升	69.14/72.78	85.18/86.64
額外樹	96.92/96.15	98.42/98.00
決策樹	96.85/95.11	98.37/97.46
SVM	1.29/1.34	50.45/50.50
邏輯回歸	5.05/4.84	52.15/52.16

III. 製造合成資料

近年來，合成資料的產生變得非常流行，尤其是在GenAI 的熱潮下。這非常有趣

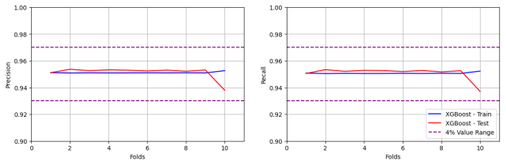


Fig. 1. XGBoost Precision-Recall Stratified 10-Fold Cross Validation.

how the generation of fake data can trick a machine learning based intrusion detection system, but even more interesting how we can use such generation techniques to construct synthetic datasets, that share statistical properties with the real ones.

A. Short Review of the Architecture of Generative Adversarial Networks and their Loss Functions

A generative adversarial network architecture is consisted of two algorithms competing each other in a minmax game [18]. The first algorithm is called Generator, whose role is to generate synthetic data identical to the real ones and is commonly noted as G. The second one is called Discriminator, whose role is to distinguish between real and synthetic data and is commonly noted as D. For the most simple architectures D and G are neural networks. Let a training data-set of dimensions $n \times p$, where n is the number of observations and p is the number of features, while each record $\mathbf{x}_i \in \mathcal{X}^p$, $i = 1, 2, ..., n$. The generator takes as input random noise, let $\mathbf{z} \in R^k$, from a k-dimensional distribution and maps it to a p-dimensional space, thus generating a new fake data-point, let $x^{(k)}_{fake} = G(\mathbf{z}) \in \mathcal{X}^p$, $k = 1, 2, ..., m$, where m is the number of newly generated data-points. Now let $D(\mathbf{x}_i) \in [0, 1]$ be the probability of the i-th data-point to be real in this binary classification task. To be able to generate new synthetic data, the generator must learn how to create data identical to the real ones. This happens through the minmax game, which is a process of optimizing the parameters of each algorithm, i.e. the weights of each neural network. Thus, given the parameters $\boldsymbol{\theta}_D, \boldsymbol{\theta}_G$ of the discriminator and the generator respectively we can notate them as $D(\mathbf{x}_i; \boldsymbol{\theta}_D)$ and $G(\mathbf{z}; \boldsymbol{\theta}_G)$ respectively [18]. The stepping-stone for optimizing the procedure is the binary cross-entropy risk function (BCE), a special occasion of the categorical cross-entropy risk function (CCE), when we address a task with two probable outcomes.

Given the discrete probability mass distributions $p(\cdot)$ and $q(\cdot)$, for a total of n data-points and C outcomes, the formulation of the risk functions of categorical cross-entropy and binary cross-entropy are formulated as seen in "(1)" and "(2)" [35], [36]:

$$R_{CCE}\Big(p(x), q(x)\Big) = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=0}^{C-1} p(x_{ij}) \log_2\Big(q(y_{ij})\Big) \quad (1)$$

$$R_{BCE}\Big(p(x), q(x)\Big) = \frac{1}{n}\sum_{i=1}^{n} p(x_{i0}) \log_2 q(y_{i0}) + \\ + \frac{1}{n}\sum_{i=1}^{n} p(x_{i1}) \log_2 q(y_{i1}) = \frac{1}{n}\sum_{i=1}^{n} p(x_{i0}) \log_2 q(y_{i0}) + \\ + \frac{1}{n}\sum_{i=1}^{n}\Big(1 - p(x_{i0})\Big) \log_2\Big(1 - q(y_{i0})\Big) \quad (2)$$

The last formulation of the binary cross-entropy is the most commonly used and can be explained since $p(y_0) + p(y_1) = 1$, where $S_Y = \{y_0, y_1\}$, where $p(\cdot)$ is a discrete probability mass distribution, with two outcomes, such that $S_Y = \{y_0, y_1\}$. From probability theory we have the following definition for the expected value of a transformation $g(X)$ of a discrete random variable X, with $f_X(x) \in [0, 1]$ as the probability mass function and $S_X = \{x_1, x_2\}$ [3]:

$$E_{f_X(x)}\Big\{g(X)\Big\} = \sum_{x \in S_X} g(x) f_X(x) = \\ = g(x_1) f_X(x_1) + g(x_2) f_X(x_2) = \\ = g(x_1) f_X(x_1) + g(x_2)\Big(1 - f_X(x_1)\Big) \quad (3)$$

Moving on from the theoretical background to the computational approach, given $\mathbf{X} \sim p(\mathbf{x})$, $\mathbf{Z} \sim q(\mathbf{z})$, where $p(\mathbf{x}), q(\mathbf{z})$ are two probability distributions, such that $p(\mathbf{x}) = 1.0$ if $\mathbf{x}$ is real, $p(\mathbf{x}) = 0.0$ if $\mathbf{x}$ is fake, one formulation of the risk function used in the Vanilla-GAN, from keras and pytorch library can be seen in "(4)" as follows [36]:

$$R_{Vanilla-GAN}\Big(D(\mathbf{x}; \boldsymbol{\theta}_D), G(\mathbf{z}; \boldsymbol{\theta}_G)\Big) = \\ = \frac{1}{n}\sum_{i=1}^{n} E_{p(\mathbf{x})}\Big\{\ln\Big(D(\mathbf{x}_i; \boldsymbol{\theta}_D)\Big)\Big\} + \\ + \frac{1}{n}\sum_{j=1}^{n} E_{q(\mathbf{z})}\Big\{\ln\Big(1 - D(G(\mathbf{z}_j; \boldsymbol{\theta}_G); \boldsymbol{\theta}_D)\Big)\Big\} = \\ = \frac{1}{n}\sum_{i=1}^{n}\Big\{p(\mathbf{x}_i)\ln\Big(D(\mathbf{x}_i; \boldsymbol{\theta}_D)\Big) + \\ + \Big(1 - p(\mathbf{x}_i)\Big)\ln\Big(D(\mathbf{x}_i; \boldsymbol{\theta}_D)\Big)\Big\} + \\ + \frac{1}{n}\sum_{j=1}^{n}\Big\{q(\mathbf{z}_j)\ln\Big(1 - D(G(\mathbf{z}_j; \boldsymbol{\theta}_G); \boldsymbol{\theta}_D)\Big) + \\ + \Big(1 - q(\mathbf{z}_j)\Big)\ln\Big(1 - D(G(\mathbf{z}_j; \boldsymbol{\theta}_G); \boldsymbol{\theta}_D)\Big)\Big\} = \\ = \frac{1}{n}\sum_{i=1}^{n}\ln\Big(D(\mathbf{x}_i; \boldsymbol{\theta}_D)\Big) + \\ + \frac{1}{n}\sum_{j=1}^{n}\ln\Big(1 - D(G(\mathbf{z}_j; \boldsymbol{\theta}_G); \boldsymbol{\theta}_D)\Big) \quad (4)$$

The minmax procedure optimizes the following risk function [7], [13], [36] and more specifically it maximizes with respect to the discriminator D and minimizes with respect to the generator G as:

$$\min_G \max_D R_{Vanilla-GAN}\Big(D(\mathbf{x}; \boldsymbol{\theta}_D), G(\mathbf{z}; \boldsymbol{\theta}_G)\Big) \quad (5)$$

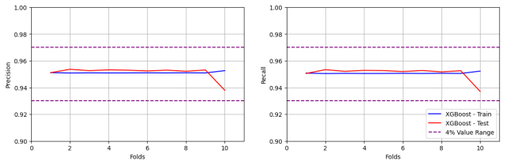


圖1. XGBoost 精確率-召回率分層10折交叉驗證。

如何偽造資料可以欺騙基於機器學習的入侵偵測系統，但更有趣的是我們如何使用這些偽造技術來構建具有與真實資料相同統計特性的合成資料集。

A. 生成對抗網絡架構及其損失函數的簡短回顧

生成對抗網絡架構由兩個在最小最大博弈中互相競爭的算法組成[18]。第一個算法稱為生成器，其角色是生成與真實數據相同的合成數據，通常記為 G。第二個算法稱為判別器，其角色是區分真實數據和合成數據，通常記為 D。對於最簡單的架構D 和 G 是神經網絡。假設一個維度為 $n \times p$ 的訓練數據集，其中 n 是觀察數量， p 是特徵數量，而每條記錄 $\mathbf{x}_i \in \mathcal{X}^p$,$i = 1\ 2\ ...\ n, , ,$ 。生成器將隨機噪聲作為輸入，讓$\mathbf{z} \in R^k$來自一個 k-維分佈，並將其映射到一個 p-維空間，從而生成一個新的假數據點，讓 $x^{(k)}_{fake} = G(\mathbf{z}) \in \mathcal{X}^p$ $k = 1\ 2\ ...\ m, , , ,$ ，其中 m 是新生成數據點的數量。現在讓 $D(\mathbf{x}_i) \in [0, 1]$ 為這個二元分類任務中第 i- 個數據點為真實的概率。為了能夠生成新的合成數據，生成器必須學會如何創造與真實數據相同的數據。這通過最小最大博弈發生，這是一個優化每個算法參數的過程，即每個神經網絡的權重。因此，給定判別器和生成器的參數$\boldsymbol{\theta}_D, \boldsymbol{\theta}_G$ ，我們可以分別記為 $D(\mathbf{x}_i; \boldsymbol{\theta}_D)$ 和 $G(\mathbf{z}; \boldsymbol{\theta}_G)$ [18]。優化程序的基石是二元交叉熵風險函數（BCE），它是分類交叉熵風險函數（CCE）的一種特殊情況，當我們處理一個具有兩種可能結果的任務時。

給定離散機率質量分佈 $p(\cdot)$ 和 $q(\cdot)$，總共有 n 個數據點和 C 個結果，分類交叉熵和二元交叉熵的風險函數的公式分別如“（1）”和“（2）”[35], [36]：

$$R_{CCE}\Big(p(x), q(x)\Big) = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=0}^{C-1} p(x_{ij}) \log_2\Big(q(y_{ij})\Big) \quad (1)$$

$$R_{BCE}\Big(p(x), q(x)\Big) = \frac{1}{n}\sum_{i=1}^{n} p(x_{i0}) \log_2 q(y_{i0}) + \\ + \frac{1}{n}\sum_{i=1}^{n} p(x_{i1}) \log_2 q(y_{i1}) = \frac{1}{n}\sum_{i=1}^{n} p(x_{i0}) \log_2 q(y_{i0}) + \\ + \frac{1}{n}\sum_{i=1}^{n}\Big(1 - p(x_{i0})\Big) \log_2\Big(1 - q(y_{i0})\Big) \quad (2)$$

二元交叉熵的最後公式是最常使用的，可以從 $p(y_0) + p(y_1) = 1$ 解釋，其中 $S_Y = \{y_0, y_1\}$，其中 $p(\cdot)$ 是一個離散機率質量分佈，具有兩個結果，使得 $S_Y = \{y_0, y_1\}$。從機率理論來看，我們對離散隨機變數 X 的變換 $g(X)$ 的期望值有以下定義，其中 $f_X(x) \in [0, 1]$ 是機率質量函數，且 $S_X = \{x_1, x_2\}$ [3]：

$$E_{f_X(x)}\Big\{g(X)\Big\} = \sum_{x \in S_X} g(x) f_X(x) = \\ = g(x_1) f_X(x_1) + g(x_2) f_X(x_2) = \\ = g(x_1) f_X(x_1) + g(x_2)\Big(1 - f_X(x_1)\Big) \quad (3)$$

從理論背景進一步到計算方法，假設 X $\sim p$(x)、Z $\sim q$(z)，其中 $p(\mathbf{x}), q(\mathbf{z})$ 是兩個機率分佈，使得 $p(\mathbf{x}) = 1.0$ 若 x 是實數， $p(\mathbf{x}) = 0.0$ 若 x 是虛數，Vanilla-GAN 中使用的風險函數的一種公式，來自 keras 和 pytorch 庫，可以在“(4)”中看到，如下 [36]：

$$R_{Vanilla-GAN}\Big(D(\mathbf{x}; \boldsymbol{\theta}_D), G(\mathbf{z}; \boldsymbol{\theta}_G)\Big) = \\ = \frac{1}{n}\sum_{i=1}^{n} E_{p(\mathbf{x})}\Big\{\ln\Big(D(\mathbf{x}_i; \boldsymbol{\theta}_D)\Big)\Big\} + \\ + \frac{1}{n}\sum_{j=1}^{n} E_{q(\mathbf{z})}\Big\{\ln\Big(1 - D(G(\mathbf{z}_j; \boldsymbol{\theta}_G); \boldsymbol{\theta}_D)\Big)\Big\} = \\ = \frac{1}{n}\sum_{i=1}^{n}\Big\{p(\mathbf{x}_i)\ln\Big(D(\mathbf{x}_i; \boldsymbol{\theta}_D)\Big) + \\ + \Big(1 - p(\mathbf{x}_i)\Big)\ln\Big(D(\mathbf{x}_i; \boldsymbol{\theta}_D)\Big)\Big\} + \\ + \frac{1}{n}\sum_{j=1}^{n}\Big\{q(\mathbf{z}_j)\ln\Big(1 - D(G(\mathbf{z}_j; \boldsymbol{\theta}_G); \boldsymbol{\theta}_D)\Big) + \\ + \Big(1 - q(\mathbf{z}_j)\Big)\ln\Big(1 - D(G(\mathbf{z}_j; \boldsymbol{\theta}_G); \boldsymbol{\theta}_D)\Big)\Big\} = \\ = \frac{1}{n}\sum_{i=1}^{n}\ln\Big(D(\mathbf{x}_i; \boldsymbol{\theta}_D)\Big) + \\ + \frac{1}{n}\sum_{j=1}^{n}\ln\Big(1 - D(G(\mathbf{z}_j; \boldsymbol{\theta}_G); \boldsymbol{\theta}_D)\Big) \quad (4)$$

The minmax 程序優化了以下風險函數[7], [13], [36] ，並更明確地說，它對於判別器 D 進行最大化，對於生成器 G 進行最小化，如下所示：

$$\min_G \max_D R_{Vanilla-GAN}\Big(D(\mathbf{x}; \boldsymbol{\theta}_D), G(\mathbf{z}; \boldsymbol{\theta}_G)\Big) \quad (5)$$

For the conditional architecture, we add a condition in the data, the target variable, which is categorical, after it is One-Hot encoded. In this architecture, the i-th input is not $\mathbf{x}_i$ as before, but has the form $(\mathbf{x}_i, \mathbf{y}_i)$, where $\mathbf{y}_i$ can be the encoded label for the i-th observation $\mathbf{x}_i$. In the basic architecture of CGAN, the minimax procedure has the following form, as seen in [36]:

$$\min_G \max_D R_{CGAN}\Big(D\big((\mathbf{x},\mathbf{y});\boldsymbol{\theta}_D\big), G\big((\mathbf{z},\mathbf{y});\boldsymbol{\theta}_G\big)\Big) \quad (6)$$

Moving to the f-GAN architectures, also known as variational divergence minimization [33], the objective function is based on the idea of the convex conjugate function $f^*(t) = \sup_{u\in dom_f}\{ut - f(u)\}$, the Fenchel conjugate, of a convex, lower-semicontinuous function $f(u) = \sup_{t\in dom_{f^*}}\{tu - f^*(t)\}$. The minimax game occurs between the generator and the discriminator, or otherwise the variational function and are noted as $D(\mathbf{x};\boldsymbol{\theta}_D)$ (or $T_{\boldsymbol{\theta}_D}(\mathbf{x}) = g_f(V_{\boldsymbol{\theta}_D}(\mathbf{x}))$) and $G(\mathbf{z};\boldsymbol{\theta}_G)$ respectively. Given an activation function $g_f(\cdot)$, related to the conjugate $f^*(\cdot)$, for $V_{\boldsymbol{\theta}_D}(\mathbf{x})$, the objective function, of the minmax process is formulated as seen is "(7)":

$$F = \Big\{E_{p(\mathbf{x})}\Big(D(\mathbf{x};\boldsymbol{\theta}_D)\Big) - E_{q(\mathbf{x})}\Big(f^*\big(G(\mathbf{z};\boldsymbol{\theta}_G)\big)\Big)\Big\} \quad (7)$$

The generator functions $f(u)$, along with their conjugate $f^*(t)$, for each f-divergence [33], which we used can be seen in Tab. III:

TABLE III
f-GAN GENERATOR FUNCTIONS AND THEIR CONJUGATES

Divergence	$f(u)$	$f^*(t)$
Kullback-Leibler	$ulog(u)$	$\exp\{t-1\}$
Squared-Hellinger	$\left(\sqrt{u}-1\right)^2$	$\frac{t}{1-t}$

B. Training Adversarial Models

For the generation of new synthetic data, we decided to select a wide range of different adversarial architectures. We chose Vanilla-GAN, Wasserstein-GAN (WGAN) [23], Wasserstein-GAN with gradient penalty (WGAN-GP) [23], f-GANs with Kullback-Leibler (KLD-GAN) and Squared-Hellinger divergence (H2D-GAN) [33], the basic architecture of Conditional-GAN for tabular data (CTGAN-1) [12], a Conditional-GAN equipped with Bayesian-Gaussian Mixture (CTGAN-2) [21], a Variational Auto-Encoder (TVAE) [26], an XGBoost-based Diffusion Forest (XGB-DF) [12], the DistilGPT2 [12] and the PATE-CTGAN, a specialized privacy-preserving conditional generative adversarial network, which utilizes the private aggregation of teacher ensembles architecture (PATE) [15].

For the training of the adversarial networks we tested a few neural network architectures, we consider a total number of 20 epochs for each of the above-mentioned algorithms. From the minimax procedure, for illustration purposes we can see below in Fig. 2 the discriminator's and generator's loss values for a few batches, for the Vanilla-GAN and WGAN, from which we observe the failure of Vanilla-GAN as the discriminator's loss converges to zero and the stable training WGAN, along with indications of convergence:

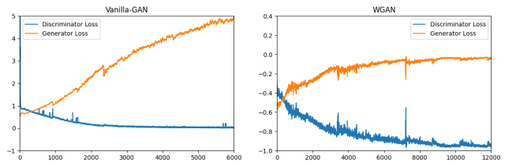


Fig. 2. GAN and WGAN Loss Progress.

Furthermore we highlight the loss progress of TVAE, whose results are the highest, compared to the rest non-conditional architectures. Although it fails generating equal number of observations per network attack. It's performance could be further optimized with the use of a conditional architecture and a mixture approach:

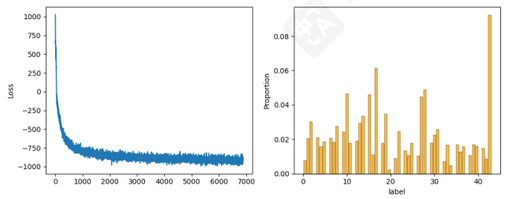


Fig. 3. TVAE Results

C. Evaluation

For evaluating the quality of the new synthetically generated data we chose the Synthetic Data Vault framework (SDV) [26]. From the quality report, we investigated the statistical fidelity, measuring how well we have captured the marginal distribution of each feature using the Kolmogorov-Smirnov Complement and how well the correlations have been preserved using the Correlation Similarity, thus leading us to a final score which is the average of the two pre-mentioned scores. We also run the diagnostic report, in order to verify if the structural integrity and data validity have been maintained with the Table Structure score, which investigates if data-types per feature between the real and the synthetic are the same and the Boundary Adherence, which checks if the minimum-maximum boundaries per feature are respected. As before with the quality report we have a final score for the diagnostic report, which is the average of the Table Structure score and the Boundary Adherence score. For the computation of the above-mentioned metrics we used only the training set. We consider the 65% and 95% thresholds for the averaged Quality and Diagnostic report scores, to select a few algorithms to proceed to further evaluation. The averaged scores can be seen in Tab. IV, where we highlight the performance of CTGAN-2 [21], the diffusion forest and the LLM [12].

After the selection of the models, we move forward with the Distinguishability Test, in order to further investigate if

對於條件式架構，我們在數據中添加一個條件，即目標變數，它是一個分類變數，在進行 One-Hot 編碼之後。在這種架構中，第 i 個輸入不是 x_i ，如之前那樣，而是具有形式 $(\mathrm{x}_i, \mathrm{y}_i)$，其中 y_i 可以是第 i 個觀察值 x_i 的編碼標籤。在 CGAN 的基本架構中，minmax 程序具有以下形式，如 [36] 所示：

$$\min_G \max_D R_{CGAN}\Big(D\big((\mathbf{x},\mathbf{y});\boldsymbol{\theta}_D\big), G\big((\mathbf{z},\mathbf{y});\boldsymbol{\theta}_G\big)\Big) \quad (6)$$

移動到 f-GAN 架構，也稱為變分分歧最小化 [33]，，目標函數基於凸共軛函數 $f^*(t)$ =$\sup_{u\in dom_f}\{ut - f(u)\}$，即一個凸、下半連續函數

$$f\ u \quad = \quad _{t\in dom_{f^*}}\{tu - (\mathrm{x}) \text{ 的 } f^*\ t\ \}$$

Fenchel 共軛，其定義為 sup_x $D(\mathrm{x};\ \boldsymbol{\theta}_D)$。在生成器與判別器之間，或變分函數與 $T_{\boldsymbol{\theta}_D}(\mathbf{x}) = g_f(V_{\boldsymbol{\theta}_D}(\mathbf{x}))$之間，發生極小極大博弈，分別標註為 $D(\mathrm{x};\ \boldsymbol{\theta}_D)$（或 $T_{\boldsymbol{\theta}_D}(\mathbf{x}) = g_f(V_{\boldsymbol{\theta}_D}(\mathbf{x}))$與 $G(\mathbf{z};\boldsymbol{\theta}_G)$）。給定一個激勵函數 $g_f(\cdot)$，與共軛 $f^*(\cdot)$ 相關，對於 $V_{\boldsymbol{\theta}_D}(\mathbf{x})$，極小極大過程的目標函數，如公式 "（7）" 所示：

$$F = \Big\{E_{p(\mathbf{x})}\Big(D(\mathbf{x};\boldsymbol{\theta}_D)\Big) - E_{q(\mathbf{x})}\Big(f^*\big(G(\mathbf{z};\boldsymbol{\theta}_G)\big)\Big)\Big\} \quad (7)$$

生成器函數 $f(u)$，及其共軛$f^*(t)$，對於我們使用的每一個 f-分歧 [33]，如表 III 所示：

表三
f-GAN 建構函數及其伴生函數

散度	$f(u)$	$f^*(t)$
Kullback-Leibler	$ulog(u)$	$\exp\{t-1\}$
平方-Hellinger	$\left(\sqrt{u}-1\right)^2$	$\frac{t}{1-t}$

B. 訓練對抗模型

為了生成新的合成數據，我們決定選擇廣泛的不同對抗架構。我們選擇了 Vanilla-GAN、Wasserstein-GAN (WGAN) [23],Wasserstein-GAN 帶梯度懲罰 (WGAN-GP) [23],f-GANs 帶 Kullback-Leibler (KLD-GAN) 和平方-Hellinger 散度 (H2D-GAN) [33], 表格式數據的基本條件-GAN 架構 (CTGAN-1) [12],一個帶有 Bayes-Gaussian 混合的條件-GAN (CTGAN-2) [21], 一個變分自編碼器 (TVAE)[26],一個基於 XGBoost 的擴散森林 (XGB-DF) [12],DistilGPT2 [12] 以及 PATE-CTGAN，一個專門的隱私保護條件生成對抗網絡，它利用了老師集體的私隱聚合架構 (PATE) [15]。

為了訓練對抗網絡，我們測試了幾種神經網絡架構，我們認為每種上述算法的總迭代次數為20次。從極小極大程序來說，為了說明目的，我們可以在圖2中看到Vanilla-GAN和WGAN的判別器和生成器的損失值，對幾個批次來說，從中我們觀察到Vanilla-GAN的失敗，因為判別器的損失收斂到零，穩定的訓練WGAN，以及收斂的指標：

損失收斂到零和穩定訓練WGAN，以及收斂的指標：

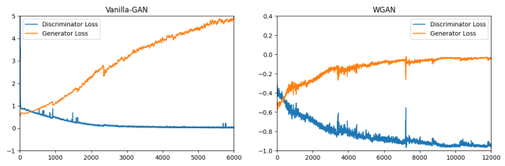


圖2. GAN與WGAN損失進度。

此外，我們突顯了TVAE的損失進度，其結果在所有非條件架構中最高。儘管它在每個網絡攻擊中無法生成相同數量的觀測值。它的性能可通過使用條件架構和混合方法進一步優化：

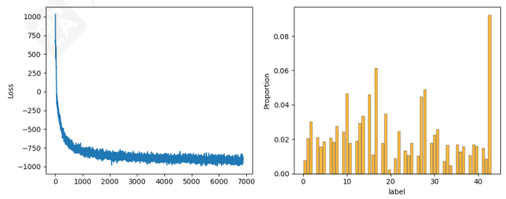


圖3. TVAE結果

C. 評估

為評估新合成數據的質量，我們選擇了合成數據保管庫框架（SDV)[26]。從質量報告中，我們調查了統計保真度，測量我們如何捕捉每個特徵的邊際分佈，使用Kolmogorov-Smirnov補充測量，以及如何保留相關性，使用相關性相似度，從而導致最終得分是前兩個得分的平均值。我們還運行了診斷報告，以驗證結構完整性和數據有效性是否得到維持，使用表結構得分調查實際數據和合成數據中每個特徵的數據類型是否相同，以及邊界遵守度，檢查每個特徵的最小-最大邊界是否得到尊重。與質量報告相同，我們為診斷報告有一個最終得分，它是表結構得分和邊界遵守得分的平均值。在上述指標的計算中，我們只使用了訓練集。我們考慮了65%和95%的質量報告和診斷報告得分的平均值，以選擇一些算法進行進一步評估。平均得分可見於表IV，我們突顯了CTGAN-2[21],、擴散森林和LLM [12]的性能。

選擇完模型後，我們進行區別性測試，以進一步調查是否

TABLE IV
SDV OVERALL SCORES

Model	*Quality Report (%)*	*Diagnostic Report (%)*
Vanilla-GAN	60.43	78.37
WGAN	57.07	81.24
WGAN-GP	63.55	78.72
KLD-GAN	63.55	78.72
H2D-GAN	49.85	80.22
CTGAN-1	75.82	83.79
CTGAN-2	98.91	100.00
TVAE	83.15	86.52
XGB-DF	99.93	100.00
DistilGPT2	99.94	100.00
PATE-CTGAN	65.00	95.17

the synthetic data mimic the statistical properties of the real data. We consider two binary classifiers, a Random-Forest and the Logistic Regression as a baseline. From the baseline model results in Tab. VI we observe that it decides randomly whether it's a real or a fake data-point since the Roc-Auc score is almost 50% and the F1 score is 0%, for the first three models. For the differentially-privacy enhanced model we see that the baseline binary classifier, can't distinguish easily but tends to realize real from fake. As seen in Tab. V we see a good F1 score, fluctuating around 52% for all models, but from the Roc-Auc we have indications that the Random Forest discriminator struggles, but slightly favors the real data, since the scores are close to 64%, which indicates that there is space for improvement. The only generative model, whose synthetic data can be easily distinguished by the random forest, is the PATE-CTGAN.

TABLE V
RANDOM FOREST - DISTINGUISHABILITY TEST RESULTS (%)

Model	*Test-F1*	*Test-Roc-Auc*
CTGAN-2	54.24	64.24
XGB-DF	53.39	65.07
DistilGPT2	52.80	63.81
PATE-CTGAN	100.00	100.00

TABLE VI
LOGISTIC REGRESSION - DISTINGUISHABILITY TEST RESULTS (%)

Model	*Test-F1*	*Test-Roc-Auc*
CTGAN-2	0.00	50.85
XGB-DF	0.13	52.80
DistilGPT2	0.20	53.70
PATE-CTGAN	14.65	63.18

To further investigate the diversity and utility of the new generated data-sets and the scenario of mode collapse, we the perform Train-Real-Test-Synthetic (TRTS) and Train-Synthetic-Test-Real (TSTR) and compare the results with the Train-Real-Test-Real (TRTR) tests. For this comparison we selected the Random Forest classifier and as evaluation metrics the precision and recall score. In Tab. VII we can find the precision and recall results for the synthetic data, which acted as a test set. From these scores we observe that they match the precision and recall scores for the original test data, apart from those of the PATE-CTGAN. From Tab. VIII we observe that the precision and recall scores of a newly trained Random-Forest, on the synthetic data, are fluctuating in the same levels of the scores seen in TRTR and TRTS. Though from PATE-CTGAN we have clear indications of poor data quality and lack of realism, for machine learning based tasks. All these results indicate high utility-quality data, with diversity and absence of mode collapse in CTGAN-2, XGB-DF and DistilGPT2.

TABLE VII
TRTS RESULTS (%)

Model	*Precision*	*Recall*
CTGAN-2	96.59	96.51
XGB-DF	96.67	96.59
DistilGPT2	96.67	96.59
PATE-CTGAN	00.54	0.02

TABLE VIII
TSTR RESULTS (%)

Model	*Train/Test Precision*	*Train/Test Recall*
CTGAN-2	96.60/95.65	96.51/95.38
XGB-DF	96.76/95.56	96.67/95.26
DistilGPT2	96.77/95.56	96.67/95.26
PATE-CTGAN	75.22/00.18	11.27/3.60

We also investigated how well each generative model has captured the distribution of the data, by comparing them with a few divergence measures [18], as seen below in Tab. IX. Almost all models achieved, at a significant level, to generate sufficient samples from each attack category, thus capturing the underlying univariate distribution of the target variable.

TABLE IX
DIVERGENCE-BASED EVALUATION METRICS

Divergence	*Jensen-Shannon*	*Hellinger*	*Wasserstein*
CTGAN-2	0.000967	0.031219	0.129161
XGB-DF	3.541914e-10	1.881997e-5	0.000106
DistilGPT2	3.541914e-10	1.881997e-5	0.000106
PATE-CTGAN	0.004467	0.066443	0.729779

For illustration purposes we show below in Fig. 4 and Fig. 5 the relative frequency per attack that was generated from CTGAN-2, the PATE-CTGAN and DistilGPT2, compared to the real proportions from the training set.

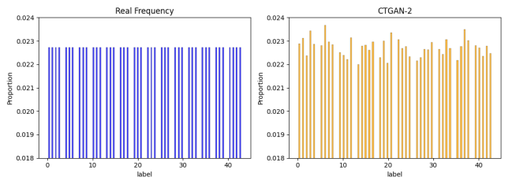


Fig. 4.

We also performed statistical hypothesis tests in order to compare the multivariate means and the covariance matrices

表IV SDV整體分數

模型	品質報告 *(%)*	診斷報告 *(%)*
Vanilla-GAN	60.43	78.37
WGAN	57.07	81.24
WGAN-GP	63.55	78.72
KLD-GAN	63.55	78.72
H2D-GAN	49.85	80.22
CTGAN-1	75.82	83.79
CTGAN-2	98.91	100.00
TVAE	83.15	86.52
XGB-DF	99.93	100.00
DistilGPT2	99.94	100.00
PATE-CTGAN	65.00	95.17

合成資料模擬了真實資料的統計特性。我們考慮兩個二元分類器，一個是隨機森林，另一個是邏輯回歸作為基線。從表 VI 的基線模型結果中，我們觀察到它在判斷是真實還是偽資料點時是隨機的，因為 Roc-Auc 分數接近 50% 且 F1 分數為 0%，對於前三個模型。對於差分隱私加強模型，我們看到基線二元分類器無法輕易區分，但傾向於識別真實資料與偽資料。從表 V 中我們看到所有模型的 F1 分數都很好，約在 52% 附近波動，但從 Roc-Auc 分數來看，隨機森林判別器表現不佳，但稍微傾向於真實資料，因為分數接近 64%，這表明有提升空間。唯一一個生成模型，其合成資料可以被隨機森林輕易區分，是 PATE-CTGAN。

表 V
隨機森林 - 區別性測試結果 (%)

模型	測試-*F1*	測試-*Roc-Auc*
CTGAN-2	54.24	64.24
XGB-DF	53.39	65.07
DistilGPT2	52.80	63.81
PATE-CTGAN	100.00	100.00

表 VI
邏輯回歸 - 判別性測試結果 (%)

模型	測試-*F1*	測試-*Roc-Auc*
CTGAN-2	0.00	50.85
XGB-DF	0.13	52.80
DistilGPT2	0.20	53.70
PATE-CTGAN	14.65	63.18

為了進一步探究新產生的數據集的多元性和實用性，以及模式崩潰的場景，我們執行 Train-Real-Test-Synthetic (TRTS) 和 Train- Synthetic-Test-Real (TSTR)，並將結果與 Train-Real-Test-Real (TRTR) 測試進行比較。為此比較，我們選擇了隨機森林分類器，並將精確度和召回率得分作為評估指標。在表 VII 中，我們可以找到用於測試集的合成數據的精確度和召回率結果。從這些得分中，我們觀察到它們與原始測試數據的精確度和召回率得分相符，除了 PATE-CTGAN 的。從表 VIII 我們觀察到，一個新訓練的隨機森林，在合成數據上，其精確度和召回率分數在波動的水平與 TRTR 和 TRTS 見到的分數相同。儘管從 PATE-CTGAN 我們有明確的指示表明數據品質差和缺乏現實性，對於基於機器學習的任務。所有這些結果都表明高效用-品質數據，在 CTGAN-2、XGB-DF 和 DistilGPT2 中具有多樣性並且沒有模式崩潰。

表 VII
TRTS 結果 (%)

模型	精確度	回想
CTGAN-2	96.59	96.51
XGB-DF	96.67	96.59
DistilGPT2	96.67	96.59
PATE-CTGAN	00.54	0.02

表 VIII TSTR 結果 (%)

模型	訓練/測試精準度	訓練/測試召回率
CTGAN-2	96.60/95.65	96.51/95.38
XGB-DF	96.76/95.56	96.67/95.26
DistilGPT2	96.77/95.56	96.67/95.26
PATE-CTGAN	75.22/00.18	11.27/3.60

我們也研究了每個生成模型如何成功地捕捉數據的分佈，透過與一些發散度量 [18], 的比較，如表IX所示。幾乎所有模型都達到了顯著水平，從每個攻擊類別中生成足夠的樣本，從而捕捉目標變量的潛在單變量分佈。

表IX
基於發散的評估指標

發散	*Jensen-Shannon*	*Hellinger*	*Wasserstein*
CTGAN-2	0.000967	0.031219	0.129161
XGB-DF	3.541914e-10	1.881997e-5	0.000106
DistilGPT2	3.541914e-10	1.881997e-5	0.000106
PATE-CTGAN	0.004467	0.066443	0.729779

為說明目的，我們在圖4和圖5中顯示了從 CTGAN-2、PATE-CTGAN和DistilGPT2生成的相對頻率每次攻擊，與訓練集的實際比例進行比較。

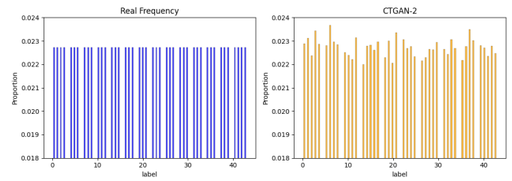


圖4。

我們也進行了統計假說檢定，以比較多變數平均值和共變數矩陣

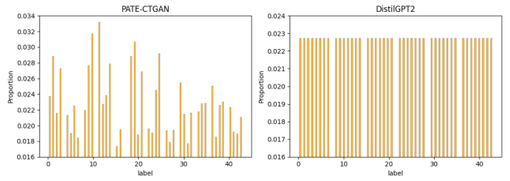


Fig. 5.

between the real and the synthetic data. We utilized the Regularized Hotelling's T^2 test [19] and the Frobenius Norm Covariance test [14], [16], for the multivariate means and covariance test respectively, each of which was computed for a total of 500 permutations, to conclude to the $p-values$ seen in Tab. X. We consider a significance level of $a = 5\%$, for both statistical tests seen in "(8)" and "(9)". Regarding the equality of means, from all the below $p-values$, as seen in Tab. X, we have evidence to reject the alternative hypothesis H_1, at a significance level $a = 0.05$, since $p-value > a$, which leads us into the conclusion that the difference between the multivariate means of the real and synthetic data are not statistically significant. From the $p-values$ of the Frobenius test, as seen in Tab. XI, we have evidence to reject the alternative hypothesis, at a significance level $a = 0.05$, since $p-value > a$, in all tests, except the one considering the CTGAN-2 model, from which we have clear indications to reject the null hypothesis H_0, at a significance level $a = 0.05$, since $p-value < a$, meaning the difference between the covariance matrices of the real and synthetic data is statistically different at a significance level of $a = 0.05$.

$$H_0 : \boldsymbol{\mu_1} = \boldsymbol{\mu_2} \quad against \quad H_1 : \boldsymbol{\mu_1} \neq \boldsymbol{\mu_2} \tag{8}$$

$$H_0 : \boldsymbol{\Sigma_1} = \boldsymbol{\Sigma_2} \quad against \quad H_1 : \boldsymbol{\Sigma_1} \neq \boldsymbol{\Sigma_2} \tag{9}$$

TABLE X
REGULARIZED HOTELLING'S T^2

Model	*p-values*
CTGAN-2	0.99
XGB-DF	1.00
DistilGPT2	1.00
PATE-CTGAN	0.99

TABLE XI
FROBENIUS NORM COVARIANCE TEST RESULTS

Model	*p-values*
CTGAN-2	0.0019
XGB-DF	0.9381
DistilGPT2	0.9261
PATE-CTGAN	0.14

We applied a two-sided non-parametric statistical test, Maximum Mean Discrepancy (MMD) [17], [4] to measure the distance between the joint distribution of the features of the real and synthetic datasets, with 500 permutations in randomized samples of the original real and synthetic datasets, which computes the mean embeddings in a Reproducing Kernel Hilbert Space (RKHS). We consider a significance level of $a = 5\%$ and their $p-values$ can be seen in Tab. XII, for each generative model. From the table, we observe that we have no evidence to reject the null hypothesis ($H_0 : P = Q$) at a significance level of $a = 5\%$, apart from the PATE-CTGAN model, which eventually failed to capture the multivariate distribution, mostly because of the differential privacy constraints. Thus, we can't assume that the difference between their distributions is statistically different for the CTGAN-2, the diffusion forest and the LLM.

TABLE XII
MMD TEST RESULTS

Model	*p-values*
CTGAN-2	0.0829
XGB-DF	1.0000
DistilGPT2	1.0000
PATE-CTGAN	0.0000

For the scenario of data leakage we use a privacy evaluation score for the generated data by each of the selected models [11], [28], to check if they are simply copying the existing records instead of synthesizing new ones, but also weight the trade-off between privacy and utility-quality. We utilized the Nearest Neighbor Distance Record score (NNDR). From Tab. XIII we observe the NNDR scores between the train and test sets compared to the synthetic datasets, to measure overfitting, while considering a threshold of 4% for the score difference between the train and the test sets. From the below scores, we have good NNDR scores, showing standard to high levels of privacy, especially for the PATE-CTGAN, but we have indications of overfitting for the other models, which don't have a privacy-preserving architecture.

TABLE XIII
NNDR RESULTS

Model	*Train-NNDR*	*Test-NNDR*
CTGAN-2	0.6136	0.7171
XGB-DF	0.6122	0.7194
DistilGPT2	0.6140	0.7193
PATE-CTGAN	0.9634	0.9731

IV. CONCLUSIONS

In this paper we clarified how efficient, in terms of speed, accuracy and stability, the tree and ensemble models are in classifying network attacks compared to other baseline models, when the network data are processed in tabular format. The conditional architectures, when combined with mixture models have outstanding performance in capturing the probability distribution function of the conditional variable, contrast to basic conditional architectures. We also highlight the diffusion and LLM models for their performance in the distinguishability tests and for capturing the underlying distributions

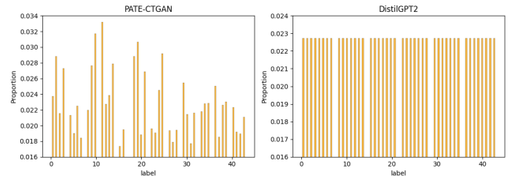


圖 5。

在真實數據和合成數據之間。我們分別利用了正則化霍特林 T^2 檢定 [19] 和 Frobenius 范數共變數檢定 [14], [16], 來進行多變數平均值和共變數檢定，每個檢定都計算了 500 次隨機排列，以得出 Tab. X 中所見的 $p-values$結果。我們對 "（8）" 和 "（9）" 中看到的兩項統計檢定均考慮了顯著性水平 $a = 5\%$。關於平均值的相等性，從所有以下 $p-values$中，如 Tab. X 所示，我們有證據拒絕替代假說 H_1，在顯著性水平 $a = 0.05$下，因為 $p-value > a$，這使我們得出結論，真實數據和合成數據的多變數平均值之間沒有統計顯著差異。從 Frobenius 檢定的 $p-values$ 中，如 Tab. XI 所示，我們有證據拒絕替代假說，在顯著性水平 $a = 0.05$下，因為$p-value > a$，在所有測試中，除了考慮 CTGAN-2 模型的測試外，我們有明確的證據拒絕零假說 H_0，在顯著性水平 $a = 0.05$下，因為 $p-value < a$，意味著真實數據和合成數據的共變數矩陣在顯著性水平 $a = 0.05$下存在統計顯著差異。

$$H_0 : \boldsymbol{\mu_1} = \boldsymbol{\mu_2} \quad against \quad H_1 : \boldsymbol{\mu_1} \neq \boldsymbol{\mu_2} \tag{8}$$

$$H_0 : \boldsymbol{\Sigma_1} = \boldsymbol{\Sigma_2} \quad against \quad H_1 : \boldsymbol{\Sigma_1} \neq \boldsymbol{\Sigma_2} \tag{9}$$

表 X
正規化霍特林 T^2

模型	*p* 值
CTGAN-2	0.99
XGB-DF	1.00
DistilGPT2	1.00
PATE-CTGAN	0.99

表 XI
弗羅貝尼烏斯規範共變異數測試結果

模型	*p*值
CTGAN-2	0.0019
XGB-DF	0.9381
DistilGPT2	0.9261
PATE-CTGAN	0.14

我們應用了一種雙邊非參數統計測試，最大平均差異 (MMD) [17], [4] 來衡量真實數據集和合成數據集特徵聯合分佈之間的距離

的距離，在隨機化的原始真實數據集和合成數據集中進行 500 次排列，它在再生核希尔伯特空間 (RKHS) 中計算平均嵌入。我們考慮了顯著性水平 $a = 5\%$ ，它們的 $p-values$ 可以在表 XII 中看到，對於每個生成模型。從表中，我們觀察到，在顯著性水平 $a = 5\%$ 下，我們沒有足夠的證據拒絕原假設 ($H_0 : P = Q$)，除了 PATE-CTGAN 模型，它最終未能捕捉到多變量分佈，主要是因為差分隱私約束。因此，我們不能假設它們之間的分佈差異在統計上是不同的，對於 CTGAN-2、擴散森林和 LLM。

表 XII
MMD 測試結果

模型	*p* 值
CTGAN-2	0.0829
XGB-DF	1.0000
DistilGPT2	1.0000
PATE-CTGAN	0.0000

在數據洩露的場景中，我們使用每個選定模型的隱私評估分數[11], [28],來檢查它們是否僅僅是複製現有記錄，而不是合成新的記錄，同時也衡量隱私與效用-品質之間的權衡。我們使用了最近鄰居距離記錄分數（NNDR）。從表 XIII 中，我們觀察了訓練集和測試集與合成數據集之間的 NNDR 分數，以測量過擬合，同時考慮訓練集和測試集之間分數差異的閾值為 4%。從下表中的分數來看，我們有良好的 NNDR 分數，顯示標準到高水平的隱私，特別是對於 PATE-CTGAN，但其他模型沒有隱私保護架構，我們有過擬合的跡象。

表 XIII
NNDR 結果

模型	*Train-NNDR*	*Test-NNDR*
CTGAN-2	0.6136	0.7171
XGB-DF	0.6122	0.7194
DistilGPT2	0.6140	0.7193
PATE-CTGAN	0.9634	0.9731

IV. 結論

在本研究中，我們釐清了當網路數據以表格格式處理時，樹模型與集成模型在分類網路攻擊方面相較於其他基準模型的效率，包括速度、準確性與穩定性。當條件架構與混合模型結合時，其表現出色於捕捉條件變數的概率分佈函數，與基本條件架構形成對比。我們也強調了擴散模型與LLM模型，因其表現優異於區別性測試中，並能捕捉底層分佈與品質隱私的取捨。

and the quality-privacy tradeoff. The use of robust and non-parametric statistical tests for analyzing the multivariate nature of the synthetic data, generated either from non-differential or differential architectures give us a solid perspective of the joint distribution of the features. Future work includes cross-dataset validation with other well known IDS datasets, using adversarial attacks for this multi-modal dataset, using conditional, privacy-preserving architectures for generative models, with Bayesian mixtures, for the unbalanced version of the UMNIDS dataset. Further evaluation is necessary with different differential-privacy architectures and their effect in privacy scores in combination with fidelity.

V. Acknowledgments

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使用強健且非參數統計測試來分析由非差異性或差異性架構產生的合成數據的多變數特性，為我們提供了關於特徵聯合分佈的堅實視角。未來工作包括使用其他知名IDS數據集進行跨數據集驗證，對此多模態數據集使用對抗性攻擊，使用條件、保護隱私的架構來進行生成模型，並搭配貝氏混合模型，針對UMNIDS數據集的不平衡版本。需要針對不同的差異性隱私架構進行進一步評估，及其與保真度結合時對隱私分數的影響。

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